

2.5.25

EE25BTECH11059 - Vaishnavi Ramkrishna Anantheertha

Question: If $\mathbf{a} = 2\hat{i} - \hat{j} - 2\hat{k}$ and $\mathbf{b} = 7\hat{i} + 2\hat{j} - 3\hat{k}$, then express $\mathbf{b}$ in the form $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$, where $\mathbf{b}_1$ is parallel to $\mathbf{a}$ and $\mathbf{b}_2$ is perpendicular to $\mathbf{a}$.

Solution:

Variable	Value
$\mathbf{a}$	$2\hat{i} - \hat{j} - 2\hat{k}$
$\mathbf{b}$	$7\hat{i} + 2\hat{j} - 3\hat{k}$

TABLE 0: Variables Used

$$\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \quad (0.1)$$

$$\mathbf{b} = \begin{pmatrix} 7 \\ 2 \\ -3 \end{pmatrix} \quad (0.2)$$

Let $\mathbf{b}_1 = k\mathbf{a}$ (where k is any scalar multiple)

$$\mathbf{b}_1 = \begin{pmatrix} 2k \\ -k \\ -2k \end{pmatrix} \quad (0.3)$$

$$\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2 \quad (0.4)$$

taking scalar product with $\mathbf{a}$ on both sides

$$\mathbf{b}\mathbf{a}^T = \mathbf{b}_1\mathbf{a}^T + \mathbf{b}_2\mathbf{a}^T \quad (0.5)$$

As $\mathbf{b}_2\mathbf{a}^T = 0$

And $\mathbf{b}_1 = k\mathbf{a}$

$$18 = 9k \quad (0.6)$$

$$k = 2 \quad (0.7)$$

$$\mathbf{b}_2 = \mathbf{b} - \mathbf{b}_1 = \begin{pmatrix} 7 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} 4 \\ -2 \\ -4 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} \quad (0.8)$$

$$\begin{pmatrix} 7 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ -4 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} \quad (0.9)$$

Therefore,

$$\mathbf{b}_1 = \begin{pmatrix} 4 \\ -2 \\ -4 \end{pmatrix} \quad (0.10)$$

$$\mathbf{b}_2 = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} \quad (0.11)$$

Refer to Figure

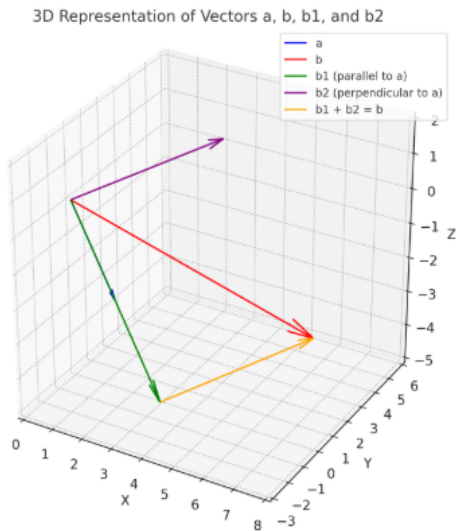


Fig. 0.1